POLITEKNIK MALAYSIA
TUN SYED NASIR



CHAPTER 2: SOURCE OF WASTES IN OIL AND FAT INDUSTRY | Ts. MUHAMMAD AZRI IZANI BIN MOHAMAD HALIM

INTRODUCTION

- There are 3 types of waste generated in oil and fats industry:

Air emission

Solid waste

Wastewater









AIR EMISSION*

- In oil and fat industry, the emissions of air pollutants can occur from a wide variety of activities during feedstock cultivation, milling and refining of the oil produced.
- Major sources of air emissions include the following:
  1. Process dust particulates.
  2. Solvent from extraction and meal processing.
  3. Odors from oils and fats manufacturing process.
  4. Boiler emissions.

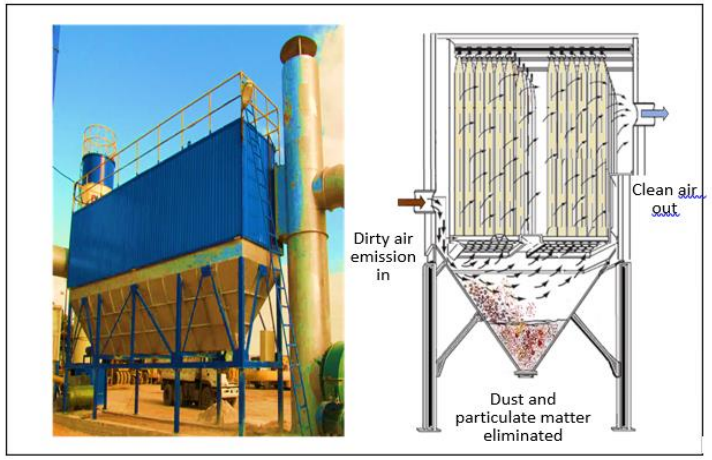
PROCESS DUST PARTICULATE

Source of air emission	Explanation	Control Measures
Process dust particulates.	Process dust particulates are produced mainly during the handling, movement, and processing of oilseeds and meal solids.	1. Good design and proper maintenance of handling systems help reduce dust during material transfer. 2. Using enclosed conveyor systems with flow monitoring helps reduce spillage and dust release. 3. Designing facilities to reduce unnecessary internal material movement also helps minimize dust generation. Installing covered conveyors and closed areas with negative pressure systems during unloading and loading prevents dust from escaping. 4. Using efficient dust control equipment such as cyclones, bag filters (baghouses), and electrostatic precipitators helps capture airborne particles. 5. Gas-cleaning systems are selected based on factors like particle size, dust concentration, flow rate, safety requirements, efficiency needed, and cost.

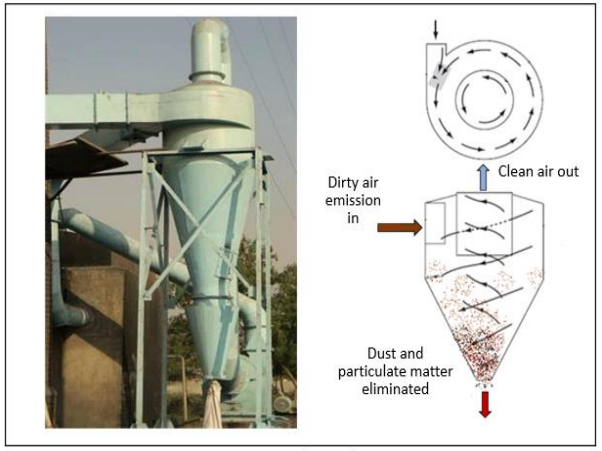
DUST CONTROL EQUIPMENTS*

Aspect	Cyclone Collector	Baghouse (Fabric Filter)	Electrostatic Precipitator (ESP)
Basic Principle	Uses spinning air (centrifugal force) to separate dust, which falls into a hopper	Air passes through fabric bags that trap dust on the surface	Uses electric charge to attract dust to metal plates
Particle Size Controlled	Large and medium particles	Very fine particles	Extremely fine and submicron particles
Efficiency	Around 90% for large particles	Almost 100% for fine dust	98%–99.9% efficiency
Advantages	Low cost, simple, low maintenance, good as pre-filter	Very high efficiency, good for fine dust	Very high efficiency, handles very small particles, low running cost
Disadvantages	Poor for fine and harmful particles	High cost, may not work well in humid or explosive dust conditions	High installation cost, large space needed

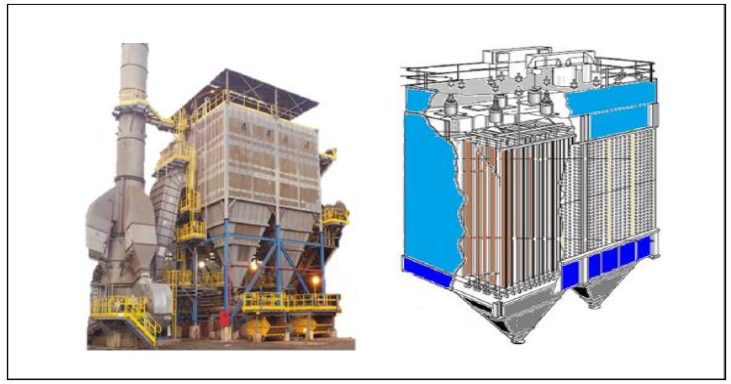
DUST CONTROL EQUIPMENTS



Bag house



Cyclone Collector



Electrostatic Precipitator (ESP)

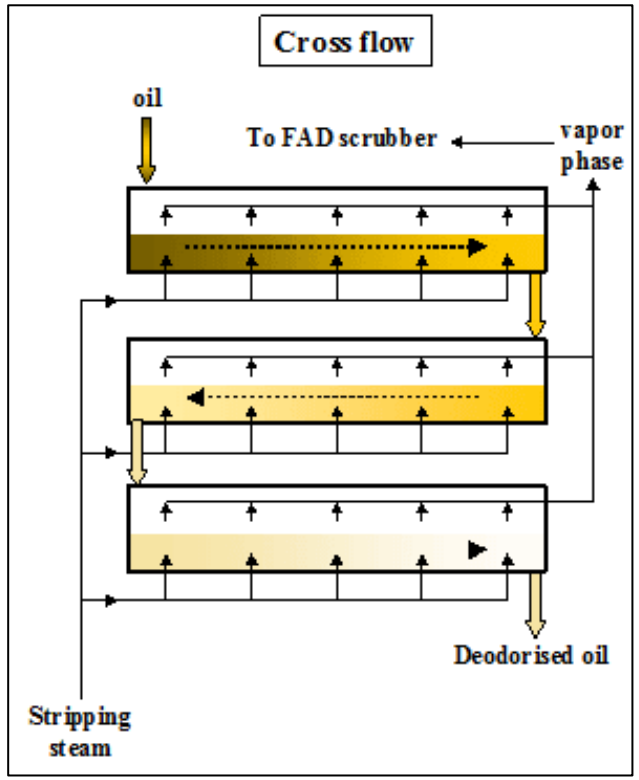
SOLVENT EXTRACTION

Source of air emission	Explanation	Control Measures
Solvent from extraction and meal processing.	Solvents (example: hexane) used to extract oil evaporate during extraction, drying, solvent recovery, storage, and handling operations.	1. Implementation of computerized monitoring and control systems to optimize process operations and improve solvent management efficiency. 2. Proper design, operation, and maintenance of distillation units, vent systems, and cooling water systems to minimize solvent evaporation and losses. 3. Effective handling, storage, and solvent recovery system to reduce solvent consumption and prevent unnecessary emissions. 4. Implementation of leak prevention system to identify and repair solvent leaks from equipment, pipelines, valves, and storage facilities.

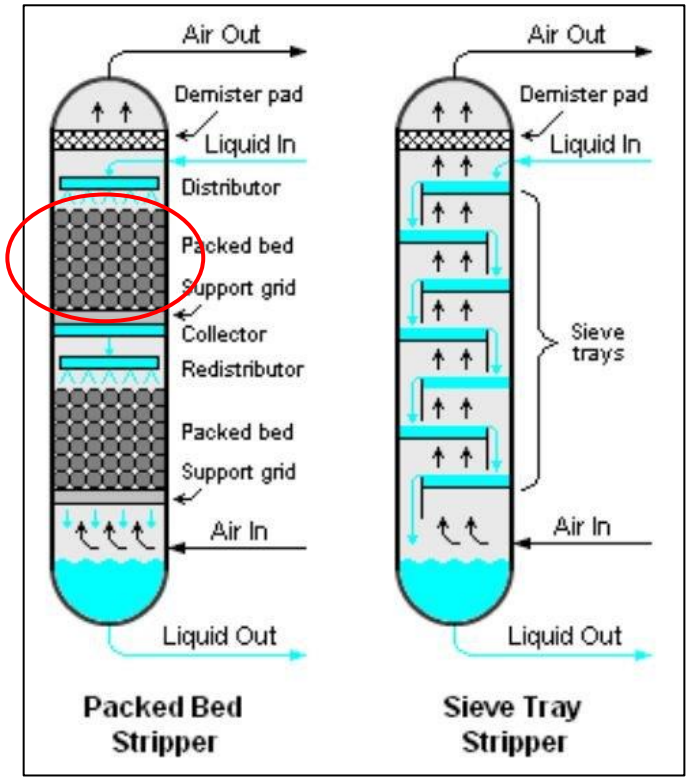
PROCESS ODOR EMISSION

Source of air emission	Explanation	Control Measures
Odors from oils and fats manufacturing process.	Odor emissions in the oil and fats industry mainly originate from **meal drying, oil deodorization, hexane residues, and acidulation processes**, where volatile organic compounds and processing vapors are released during heating, cooling, storage, and chemical treatment operations.	1. Improve deodorizers tray design according to the cross-flow principle to give good agitation thus ensuring a high overall deodorization efficiency 2. Install a packed column stripper to increase contact surface area of oil and stripping steam in a short contact time. 3. Install an additional demister ahead of the vacuum unit to minimize carryover of fatty matter to the barometric condenser. 4. Use double scrubber system for maximum condensation and recovery of fatty matter from the vapor phase with a minimum pressure drop.

ODOR EMISSION CONTROL

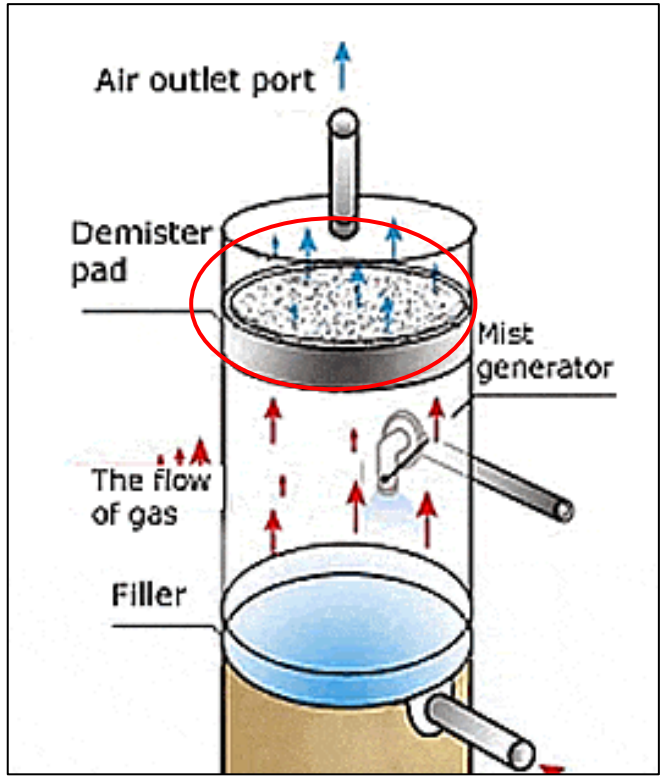


Improve deodorizers tray design according to the cross-flow principle.

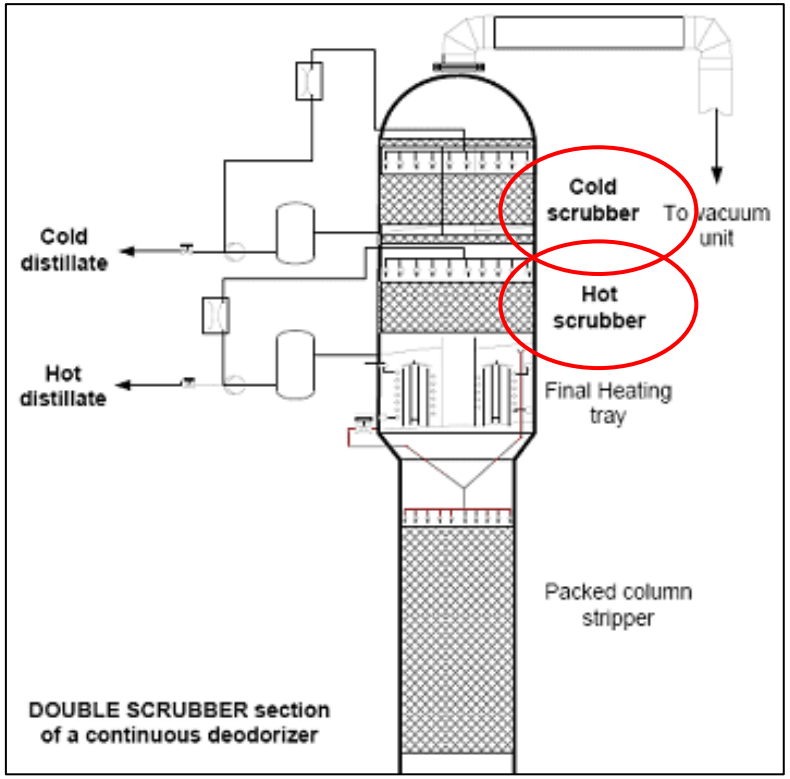


Install a packed column stripper to increase contact surface area.

ODOR EMISSION CONTROL



Install an additional demister ahead of the vacuum unit.



Use double scrubber system for maximum condensation and recovery.

BOILER EMISSION

Source of air emission	Explanation	Control Measures
Boiler emission	Boilers burn fuels to produce steam and energy, but this combustion releases harmful gases and particles such as Nitrogen Oxide (Nox), Sulphur Oxide (Sox), Carbon Monoxide (CO), and particulate matter. These pollutants reduce air quality, harm human health, damage plants, and contribute to environmental problems like acid rain and global warming.	Fuel-Fired Boiler: 1. NOx & CO control: In fuel-fired boilers, part of the combustion gases is recirculated to the burner to control combustion temperature and reduce NOx formation. Coal-Fired Boiler: 1. Particulate matter control: Bag house is the equipment to capture and remove fine particulate matter from flue gases before release to the atmosphere. 2. NOx & CO control: Fluidized bed boilers design reduce NOx and CO emissions through lower combustion temperatures and improved air-fuel mixing, while limestone in the bed captures sulfur compounds. 3. SOx control: Using low sulphur-coal entering the combustion process, thereby lowering SOx emissions.

BOILER EMISSION*

Source of air emission	Explanation	Control Measures (Continue)
Boiler emission	Boilers burn fuels to produce steam and energy, but this combustion releases harmful gases and particles such as Nitrogen Oxide (Nox), Sulphur Oxide (Sox), Carbon Monoxide (CO), and particulate matter. These pollutants reduce air quality, harm human health, damage plants, and contribute to environmental problems like acid rain and global warming.	Coal-Fired Boiler (Continue): 4. Trace metal control: Using cleaner coal with lower trace metal content reduces metal emissions. 5. Trace metal control: Air pollution control devices remove metal-containing particles from flue gas. 6. HCl & trace metal control: Alkaline sorbents (example: hydrated lime) react with HCl and remove it from flue gases.

EXAMPLE 1

1. State 4 major sources of air emission.
2. State 3 control measures of air emission from boiler.

SOLID WASTE

- Solid wastes in the oil and fat industry are mainly generated during plantation activities, oil extraction processes, and oil refining operations. These wastes may contain organic materials, spent processing materials, and used catalysts that require proper management and disposal.
- Example of solid wastes:
    1. Biomasses (fibre, shell, hulls, leaves, wood, trunk etc)
    2. Meal and grain solid / oilseed cake
    3. Refinery bleaching clay/ bleaching earth/ activated clay
    4. Spent hydrogenation nickel catalyst

SOLID WASTE

- Major sources of solid waste include the following:
    1. Plantation and deforestation
    2. Extraction of oilseed (mechanical or chemical)
    3. Oil bleaching
    4. Oil hydrogenation











DEFORESTATION & PLANTATION

Source of solid waste	Explanation	Control Measures
Deforestation & plantation	Deforestation for plantation development can lead to habitat loss, forest degradation, biodiversity reduction, and environmental pollution if not properly managed. The major solid wastes generated are biomass residues from harvesting activities, including wood, leaves, trunks, fibres, and other plant materials.	1. Fiber & shell: Fibres and shells are burned in mill boilers to generate steam and electricity, reducing reliance on fossil fuels and improving energy efficiency. 2. Old stems, leaves, fibre: These materials are spread on plantation soil as mulch to improve soil quality, retain moisture, suppress weed growth, and regulate soil temperature. As they decompose, they enhance soil structure and nutrient content. 3. Empty fruit bunch: EFB is decomposed and returned to the plantation as organic fertilizer. It improves soil fertility, increases water retention, and enhances nutrient availability for oil palm growth.

OILSEED EXTRACTION

Source of solid waste	Explanation	Control Measures
Oilseed extraction.	During the oilseed extraction process, several operations generate solid wastes as by-products. These wastes are mainly biomass materials and extraction residues that remain after oil has been removed from the oilseed.	1. Oilseed meal / cake is rich in protein, minerals, and other nutrients, making it a valuable feed ingredient for livestock such as cattle, poultry, sheep, and goats. Reusing oilseed meal as animal feed reduces solid waste generation and converts a by-product into a useful resource.

Example of sunflower meal & rapeseed meal as animal feed.





OIL BLEACHING

Source of solid waste	Explanation	Control Measures
Oil Bleaching	Bleaching is an oil refining process that uses bleaching earth to remove pigments such as carotene and chlorophyll, along with impurities like metals, soaps, and oxidation products. After use, it becomes Spent Bleaching Earth (SBE), containing 20–40% residual oil and contaminants, which is disposed of at high landfill cost.	1. Use in poultry feed (0.5 - 2.0%): Small amounts of bleaching earth can be used as a pellet binder in poultry feed formulation, helping to improve feed texture and stability. 2. Sanitary landfill disposal (with precautions): SBE can be disposed in sanitary landfills, but it poses a risk of spontaneous combustion due to heat from oxidation of residual organics. 3. Fire prevention methods: To reduce fire risk, antioxidants, water sprays, and lime are added to stabilize the waste and prevent self-heating and ignition. 4. Re-use as adsorbent: De-oiled SBE can be reused to remove contaminants in wastewater treatment, acting as a low-cost adsorbent material. 5. Construction materials: SBE can substitute clay in bricks, blocks, and tiles, reducing raw material demand in the construction industry.

OIL HYDROGENATION

Source of solid waste	Explanation	Control Measures
Oil Hydrogenation	Hydrogenation in the oil and fat industry produces solid waste mainly in the form of spent nickel catalyst. This waste arises from catalyst deactivation during hydrogenation and may pose environmental risks due to nickel content, requiring careful handling, recovery, or disposal.	1. Sanitary landfill disposal (with precautions): When recycling is not possible, spent catalyst is disposed of in sanitary landfills under controlled conditions to reduce environmental risk. 2. Catalyst recycling: The nickel catalyst is reused multiple times in hydrogenation until it becomes inefficient, reducing the amount of waste generated. 3. Nickel recovery: Spent catalyst is processed to recover nickel metal. However, recovery systems are not widely established and are only available in some regions or country.

EXAMPLE 2

1. Describe 3 control measures in handling solid waste produced from hydrogenation.

WASTEWATER*

- Wastewater in the oil and fat industry is generated from almost all processing activities and contains oil, grease, organic matter, and chemical residues.
- Its characteristics and quantity vary according to the production process, making proper treatment essential to prevent environmental pollution.
- Major sources of wastewater include the following:
    1. Oil seed plantation
    2. Oil seed preparation
    3. Oil seed extraction
    4. Crude oil refining
    5. Refined oil modification

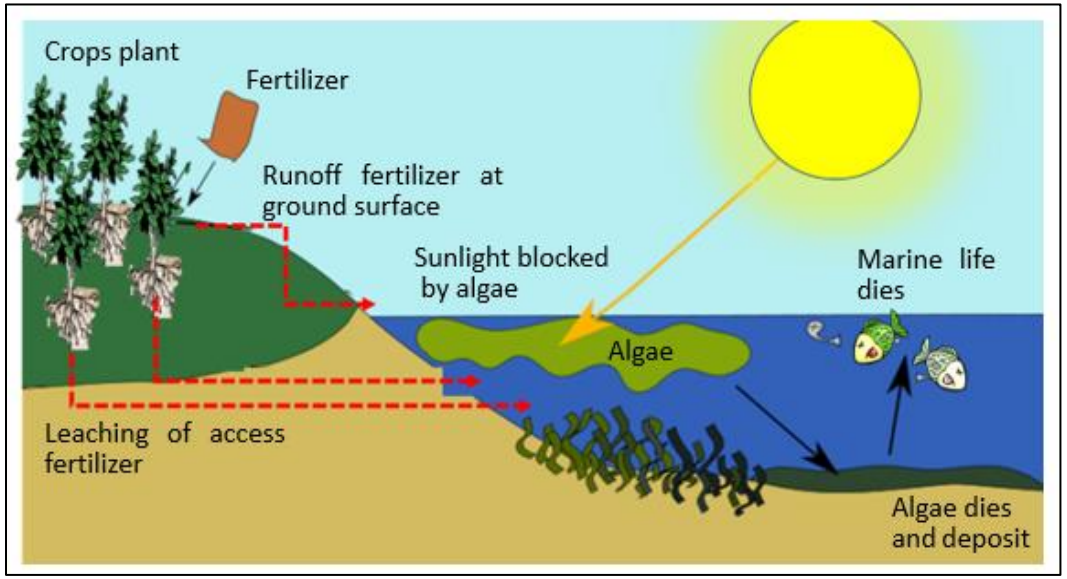
OILSEED PLANTATION

- Impact of oilseed plantation:

No	Impact of oilseed plantation	How it creates wastewater
1	Deforestation and degradation of the forest systems.	Soil disturbance increases surface runoff, carrying sediments into water bodies, forming turbid wastewater.
2	Loss of habitats including plants and animals	Ecosystem imbalance reduces natural filtration, allowing more pollutants to enter water systems.
3	Extreme soil degradation and pollution	Eroded soil particles are washed into water, increasing suspended solids in wastewater.
4	Usage of large quantities of pesticides and herbicides to maintain the plantation.	Chemicals dissolve or attach to runoff water, creating toxic agricultural wastewater.
5	Eutrophication in aquatic ecosystem	Excess nitrogen and phosphorus are washed into water, forming nutrient-rich wastewater that causes eutrophication.

EUTROPHICATION

- **How Eutrophication Happens**
- Step 1: Fertilizers from farms are washed into rivers and lakes by rainwater. These fertilizers contain extra nutrients like nitrogen and phosphorus.
- Step 2: The extra nutrients cause plants and algae in the water to grow very fast.
- Step 3: This rapid algae growth forms a thick layer on the water surface called an algal bloom, which blocks sunlight from reaching underwater plants.
- Step 4: Without sunlight, underwater plants die. When they die, bacteria start breaking them down and use up oxygen in the water.
- Step 5: Oxygen levels in the water become very low, so fish and other aquatic animals cannot survive and eventually die.



OILSEED PREPARATION

- Impact of oilseed preparation:

No	Impact of oilseed preparation	How it creates wastewater
1	Water pollution after cleaning process	Washwater carries soil, organic matter, and suspended solids into drains or rivers if untreated, increasing water contamination.
2	Thermal pollution and oily wastewater after sterilization process	Hot condensate and process water contain oil residues and organic compounds, which reduce water quality and harm aquatic life.
3	Increased organic pollution load after sterilization process	Broken plant tissues release oil and organic matter into wastewater, increasing BOD and COD levels in water systems.

OILSEED EXTRACTION

- Impact of oilseed extraction:

No	Impact of oilseed extraction	How it creates wastewater
1	Water pollution (oil contamination)	Oil residues from extraction process form a layer on water surface, blocking oxygen transfer and harming aquatic life.
2	Eutrophication risk (soybean oil)	Trace phosphorus from phosphatides acts as a nutrient, causing excessive algae growth in rivers and lakes.
3	High BOD and COD	Organic matter in wastewater increases oxygen demand, reducing oxygen available for fish and other organisms.

EXAMPLE 3

1. Explain in detail the impact of eutrophication process to aquatic life.

CRUDE OIL REFINING

- Impact of crude oil refining :

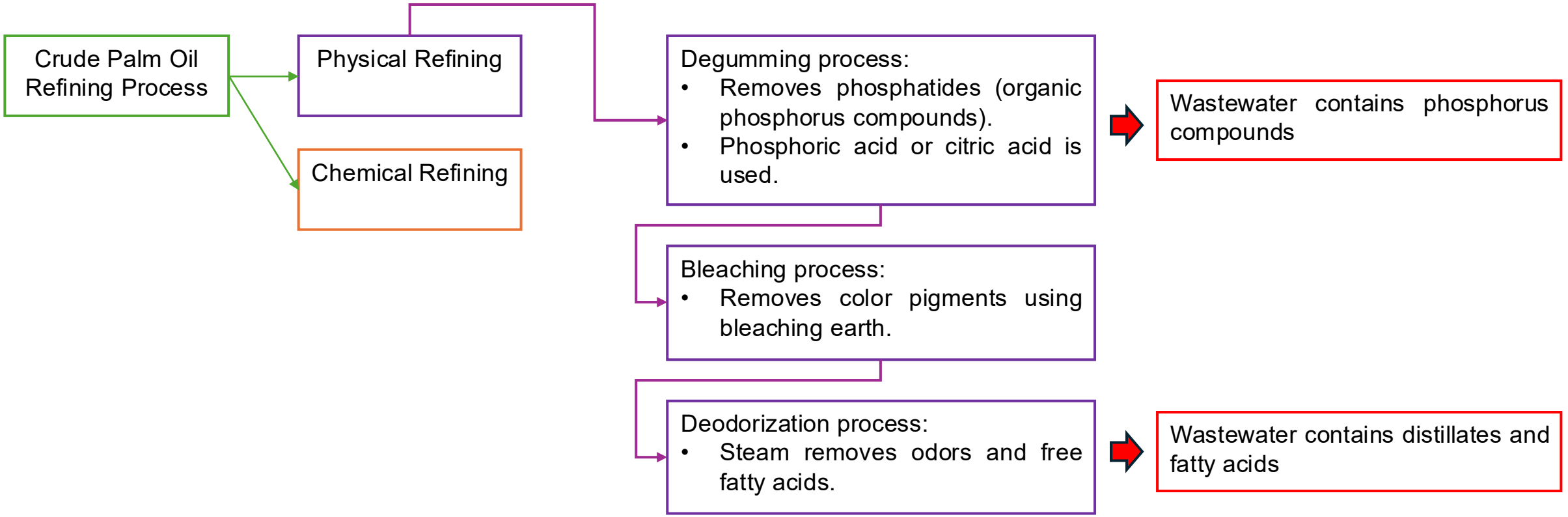
No	Impact of crude oil refining	How it creates wastewater
1	Causes eutrophication and algal blooms in water bodies	During degumming, phosphoric acid removes phosphatides (organic phosphorus compounds) from crude oil. These phosphorus compounds are transferred into wastewater. When discharged into rivers/lakes, phosphorus acts as a nutrient source for algae.
2	Increases dissolved solids and may produce odor-causing compound under anaerobic conditions	In soapstock acidulation, sulfuric acid is used, producing sulfate-rich effluent. Neutralization and washing processes add dissolved salts, soaps, and fatty residues. These increase total dissolved solids (TDS) in wastewater.

CRUDE OIL REFINING

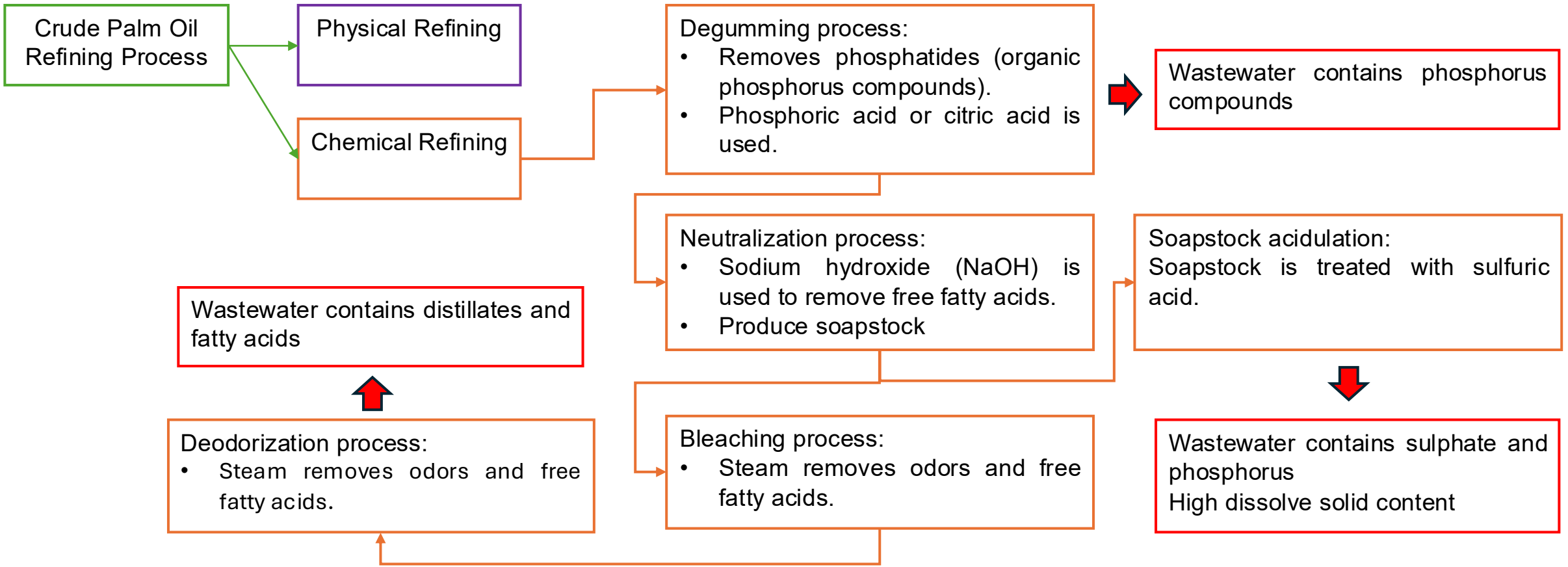
- Impact of crude oil refining (continue):

No	Impact of crude oil refining	How it creates wastewater
3	Increase BOD and COD, reducing oxygen levels in water	Extraction and refining release oil residues, phospholipids, fatty acids, and organic matter into water. These substances require oxygen for microbial decomposition.
4	Contribute to water pollution and foaming in rivers	Neutralization process produces soapstock (soap-like substances). These soaps and surfactants enter wastewater streams. Residual oils from extraction also mix with water.

CRUDE OIL REFINING



CRUDE OIL REFINING



EXAMPLE 4

1. Explain the degumming, neutralization and bleaching process in chemical crude oil refining process.

THANK YOU

PLEASE SUBMIT YOUR MIND MAP